Artificial Intelligence and Computer Vision Approaches for Apple Leaf Disease Detection and

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***Abstract*—Apple is among the most economically significant tree fruits worldwide, yet its production is persistently threatened by foliar diseases such as scab, black rot, cedar/apple rust, frog-eye leaf spot, and complex multi-disease infections. Manual, expert-based diagnosis is slow, subjective, and difficult to scale, motivating three decades of research into automated, image-based detection systems. This review synthesizes ten recent studies spanning classical machine learning, convolutional neural networks (CNNs), generative-adversarial data augmentation, hybrid metaheuristic-optimized architectures, object-detection frameworks, vision transformers (ViTs), and sensor-efficient sparse-coding techniques. We organize the literature by methodological family, summarize datasets and reported performance, and compare accuracy against computational cost (parameters, FLOPs, inference time), which is the central trade-off for real-world, in-field deployment. Reported accuracies across the reviewed works range from roughly 77% for feature-engineered SVM pipelines under natural background conditions to above 99% for deep transfer-learning and GAN-augmented CNN models on curated datasets, illustrating both the promise and the dataset-dependence of current systems. We close by identifying recurring gaps — limited field-condition validation, inconsistent dataset provenance, shallow interpretability, and the accuracy/lightweight trade-off — and outline directions for future work, including edge-optimized hybrid CNN-ViT designs, explainable AI, and IoT-integrated deployment.**

***Index Terms*—apple leaf disease, deep learning, convolutional neural network, vision transformer, precision agriculture, plant disease classification, computer vision**

I. Introduction

Apples rank among the most widely cultivated and consumed fruits globally, valued both for their nutritional profile (fiber, antioxidants, vitamin C) and for the substantial agricultural income they generate in producing regions. Fungal and bacterial pathogens—most notably *Venturia inaequalis* (scab), *Botryosphaeria obtusa* (black rot), and *Gymnosporangium juniperi-virginianae* (cedar/apple rust)—routinely damage foliage and fruit, and can cause severe yield and quality losses if not identified early. Traditional disease diagnosis relies on visual inspection by agronomists or trained farmers, a process that is labor-intensive, inconsistent across observers, and often too slow to intervene before a disease spreads through an orchard.

Over the past decade, computer vision and artificial intelligence (AI) have emerged as the dominant paradigm for automating this diagnostic task. Early work relied on hand-crafted color, texture, and shape features paired with classical classifiers such as support vector machines (SVMs). The field then shifted toward deep convolutional neural networks (CNNs), which learn discriminative features directly from raw images, followed most recently by vision transformers (ViTs) and hybrid CNN–ViT architectures that combine local feature extraction with global self-attention. In parallel, a persistent secondary research thread has focused on making these models *lightweight* enough for deployment on smartphones, drones, or embedded orchard sensors, since high accuracy is of limited practical value if a model cannot run in real time on farmer-accessible hardware.

This review synthesizes ten studies (2021–2025) that collectively represent this trajectory: from classical feature-based pipelines, through mainstream and lightweight CNNs, generative data augmentation, capsule-network/metaheuristic hybrids, YOLO-based object detectors, sparse-coding sensor pipelines, to state-of-the-art lightweight vision transformers—as well as a recent systematic review covering 45 additional studies (2016–2024) that contextualizes the broader field.

II. Review Scope and Methodology

The ten primary sources analyzed here were selected to represent the major methodological families active in apple leaf (and, in one case, apple fruit) disease detection research:

1) **Classical/hybrid feature-based ML**—ant colony optimization (ACO) feature selection with SVM classification [7].
2) **Lightweight CNN design**—a custom three-layer CNN benchmarked against nine pretrained architectures [5].
3) **Generative data augmentation**—DCGAN-based synthetic image generation combined with a compact CNN [6].
4) **Metaheuristic-optimized hybrid deep learning**—Capsule Network (CapsNet) features optimized via Water Wave Optimization (WWO) and classified with a Bidirectional LSTM [2].
5) **Object detection**—an improved YOLOv5 variant (A-Net) using Wise-IoU loss and RepVGG convolution blocks [4].

6) **Sensor-efficient classical ML**—sparse coding combined with a shallow artificial neural network for apple pest and disease detection from fruit images [8].
7) **Large-scale transfer-learning benchmarking**—twelve pretrained CNNs evaluated head-to-head [9].
8) **Hybrid CNN–ViT architectures**—AppViT, combining convolutional stem blocks with stacked transformer encoders [1].
9) **Lightweight, interpretable ViT**—AppleViT, incorporating Layer-wise Relevance Propagation (LRP) and attention visualization for explainability [3].
10) **Systematic literature review**—a PRISMA-based synthesis of 45 apple-leaf-disease deep learning studies from 2016–2024, covering datasets, frameworks, and open challenges [10].

Each is discussed in the sections that follow, ending with a cross-cutting comparative analysis.

III. Datasets Used Across the Literature

Dataset choice strongly shapes reported performance and generalizability. Across the reviewed works, three dataset families dominate:

PlantVillage [11] is the most widely used public repository (54,000+ images across 14 species and 38 classes), typically captured against plain backgrounds under controlled lighting. It is used, in various apple-specific subsets, by Vishnoi et al. (3,171 images; scab, black rot, cedar rust, healthy) [5], Fraiwan et al. (the same 3,171-image apple subset) [9], Al-Wesabi et al. (a related four-class Kaggle-hosted variant) [2], and Li et al. (a PlantVillage subset for their ACO-SVM pipeline) [7].

Plant Pathology 2021–FGVC8 [12] is a more challenging benchmark of ∼16,000–18,000 orchard-collected images with complex natural backgrounds, multiple simultaneous diseases (e.g., a "complex" class), and five to six disease categories. It is used by Ullah et al. (AppViT) as the primary training set [1] and by Gupta & Jadon (AppleViT) as an external generalization test [3].

Small or self-collected datasets are common where public benchmarks do not fit the target task: Ayaz et al. used only 319 original fruit images (blotch, rot, scab, healthy) prior to DCGAN augmentation to 4,000 synthetic images [6]; Liu and Li assembled a 12,000-image custom dataset combining real orchard photographs and a public repository for five diseases plus healthy leaves [4]; Abbaspour-Gilandeh et al. photographed 273 physical apple fruit samples (819 images across three camera views) under a controlled light chamber for pest/disease classification [8].

The systematic review by Doutoum and Tugrul further quantifies this pattern: across 45 surveyed studies, self-constructed datasets were the single largest category (∼36%), followed by PlantVillage, with the Plant Pathology dataset and specialized institutional datasets used less frequently [10]. This fragmentation—different disease taxonomies, background conditions, and image counts—is a recurring obstacle to fair cross-study comparison.

IV. Classical and Feature-Engineered Machine Learning Approaches

Li et al. [7] present a representative modern example of the "handcrafted feature" paradigm, applied to three visually similar diseases (Black Spot, Black Rot, Cedar Rust) plus healthy leaves from PlantVillage. Their pipeline performs background removal via green-channel extraction and morphological operations, isolates diseased regions using color-based K-means clustering, and extracts 68 candidate features spanning four color spaces (RGB, HSV, L*a*b*, YCbCr), gray-level co-occurrence matrix (GLCM) texture descriptors, and shape metrics (lesion count, area, perimeter, circularity). An ant colony optimization algorithm then selects the six most discriminative features per group before classification with a linear SVM.

The results illustrate how heavily preprocessing affects outcomes for feature-based methods: classification accuracy rose from 77.5% on raw images, to 83.1% after background removal, to 92.5% when the classifier was restricted to disease-specific lesion regions. Texture features (entropy, energy, skewness) proved most discriminative (80.2% standalone accuracy), followed by color (75.6%) and shape (63%). The authors argue that such interpretable, low-data-requirement pipelines remain valuable where large labeled datasets or GPU resources are unavailable, and they generalize the discussion to visually similar diseases across other crops (grape, tomato, potato, maize), underscoring that lesion ambiguity is a cross-species challenge rather than an apple-specific one.

Abbaspour-Gilandeh et al. [8] take a related but distinct approach, targeting apple *fruit* pests (apple capsid, codling moth, pear lace bug) and one physiological disorder (russeting) rather than leaf disease. Their method divides each fruit image into 64×64 sub-images, extracts eleven color/morphological features (RGB, L*a*b*, HSV, SIFT, Harris corner descriptors) per patch, and uses sparse coding to build a compact "dictionary" of representative patches before classification with a small multilayer perceptron. This sub-image strategy is explicitly aimed at reducing computational load: after ∼220 seconds to build the dictionary, per-image inference took only 0.175 seconds. Average clustering accuracy across 18 fine-grained classes reached about 90–91% with an optimally tuned single hidden layer (10 neurons, learning rate 0.2), though performance varied considerably by class (67–100%), reflecting the difficulty of discriminating co-occurring pest/disease combinations from limited per-class samples.

V. Convolutional Neural Network Approaches

A. Lightweight custom CNNs

Vishnoi et al. [5] designed a compact three-convolutional-layer CNN (Conv-3 DCNN) explicitly to reduce the computational burden associated with deep pretrained networks. Trained on 3,171 PlantVillage apple images (scab, black rot, cedar rust, healthy) with standard augmentation (shift, shear, zoom, flip) and hyperparameters tuned via random search, the model achieved 98% overall accuracy—comparable to, and in

several respects exceeding, nine pretrained transfer-learning baselines (VGG-16/19, ResNet-50/152, DenseNet-201, MobileNetV2, InceptionV3, Xception, MobileNet) evaluated on the identical split. Critically, the proposed model required substantially less storage and training/inference time than the pretrained alternatives, reinforcing its suitability for handheld or resource-constrained deployment.

B. Generative data augmentation for small datasets

Ayaz et al. [6] addressed a common bottleneck—very small original datasets—using a Deep Convolutional Generative Adversarial Network (DCGAN) to synthesize additional training images. Starting from only 319 real fruit images (Blotch, Rot, Scab, Healthy), their DCGAN generated 4,000 synthetic images (1,000 per class) to train a five-convolutional-layer DCNN. Using an Adam optimizer, the DCGAN-DCNN model achieved 99.99% overall accuracy, outperforming ResNet50, SqueezeNet, and MiniVGGNet baselines as well as several classical handcrafted-feature/MSVM pipelines from prior literature (which topped out around 95–96%). While such near-perfect accuracy warrants caution regarding overfitting to a narrow, low-resolution (32×32) synthetic distribution, the study demonstrates GAN-based augmentation as a viable strategy when real-world data collection is constrained.

C. Large-scale transfer-learning comparison

Fraiwan, Faouri, and Khasawneh [9] provide the most extensive head-to-head CNN benchmark among the reviewed works, fine-tuning twelve well-established pretrained architectures (DarkNet-53, DenseNet-201, EfficientNet-b0, GoogLeNet, Inception-v3, Inception-ResNet-v2, MobileNetV2, ResNet-18/101, ShuffleNet, SqueezeNet, Xception) on the same 3,171-image apple PlantVillage subset used by Vishnoi et al. Under both 50/50 and 70/30 train/test splits, nearly all models achieved F1 scores, precision, recall, specificity, and accuracy above 99%, with Inception-v3, DenseNet-201, and MobileNetV2 achieving near-perfect or perfect scores at the 70/30 split. The weakest performer, Inception-ResNet-v2, still achieved a 96.5% F1 score but showed signs of overfitting to the larger training partition. Scab images were the most frequent source of residual confusion (often misclassified as healthy). The authors also report training/validation time, revealing an order-of-magnitude spread (as low as $\sim$119 s for SqueezeNet versus over 4,200 s for Xception at the larger data split)—a practical consideration largely orthogonal to raw accuracy but decisive for deployment feasibility.

VI. Metaheuristic-Optimized Hybrid Deep Learning

Al-Wesabi et al. [2] propose an "AI-enabled apple leaf disease classification" (AIE-ALDC) pipeline that fuses several distinct components: orientation-based data augmentation (image rotation and mirroring), Gaussian-filter noise removal, Capsule Network (CapsNet)-based feature extraction (using vector "capsules" and dynamic routing rather than scalar activations), Water Wave Optimization (WWO)—a nature-inspired metaheuristic modeled on ocean-wave propagation, refraction, and breaking—to tune CapsNet hyperparameters, and a Bidirectional LSTM (BiLSTM) classifier to assign final disease labels. Evaluated on a four-class dataset (Apple Scab, Black Rot, Cedar Rust, Healthy; $\sim$2,000 training and $\sim$500 test images per class), the model achieved an average accuracy of 99.2% across five experimental runs, with precision, recall, specificity, and F-score all in the 97–99.8% range. Comparative analysis showed AIE-ALDC requiring less memory (2.56 GB) and training time (0.518 h) than AlexNet, GoogLeNet, VGGNet-16, ResNet-20, and a standard DCNN baseline, while also outperforming SVM, backpropagation networks, and several ResNet variants (accuracy range 55–97% in prior literature) on classification accuracy. This work exemplifies a broader trend of combining biologically inspired optimization with modern feature-extraction architectures to squeeze additional accuracy and efficiency from a fixed model budget.

VII. Object Detection Approaches

Whereas the previous methods perform whole-image classification, Liu and Li [4] target the more operationally useful task of detecting and localizing individual disease lesions within a leaf image—important when leaves display multiple simultaneous infections or when disease severity mapping is needed. Their A-Net model modifies YOLOv5-6.0 in two ways: replacing the standard CIoU loss with **Wise-IoU**, which incorporates an attention mechanism and dynamic focusing to down-weight low-quality anchor boxes and accelerate convergence, and substituting the original convolutional backbone with **RepVGG** blocks, which use structural re-parameterization to fuse multi-branch training-time convolutions into a single efficient branch at inference time. On a custom 12,000-image dataset covering five diseases (rust, mosaic, black rot, brown spot, gray spot) plus healthy leaves, ablation experiments showed each modification improving precision and recall independently, with the combined model reaching 90.7% precision, 93.8% recall, and 94.8% mAP@0.5—improvements of 3.9, 4.2, and 5.2 percentage points respectively over the unmodified YOLOv5 baseline, and clear margins over SSD and Faster R-CNN comparators. Data augmentation (rotation, translation, and mixed lab/field backgrounds) further improved mAP from 73.1% to 79.1%. The authors note that despite these gains, training time remained a limitation, and they identify further model compression for drone/mobile deployment as future work.

VIII. Vision Transformers and Hybrid CNN–ViT Architectures

Two of the reviewed studies represent the most recent methodological shift toward transformer-based architectures, motivated by ViTs' ability to model long-range spatial dependencies via multi-head self-attention (MHSA)—a capability CNNs lack due to their local receptive fields.

AppViT [1] is a hybrid architecture that alternates convolutional blocks (for local feature encoding and progressive downsampling) with stacked ViT encoder blocks (for global context modeling), followed by global average pooling and a softmax classifier. Trained on the challenging Plant Pathology 2021–FGVC8 dataset (five disease classes plus healthy, expanded via augmentation from 16,093 to 44,507 images), AppViT achieved 96.4% accuracy with only about 1.3 million parameters and 0.644 billion FLOPs—substantially fewer resources than VGGNet (138M parameters, 15.3 GFLOPs), ResNet-50/152, Inception-V3, and even lightweight EfficientNet-B3/B4 variants, while simultaneously outperforming all of them in accuracy, precision (0.967), recall (0.959), and F1-score (0.963). This result is notable because it directly contradicts the common assumption that lightweight models must sacrifice accuracy; AppViT outperformed EfficientNet-B3/B4 by 4.3 and 5 percentage points respectively while using roughly 85–93% fewer parameters and FLOPs.

AppleViT [3] extends this hybrid design philosophy with an explicit focus on interpretability and field-readiness. Built around standard ViT components—patch embedding, stacked transformer encoder blocks with MHSA and feed-forward networks, and a classification head—the $\sim$1.3M-parameter model was trained on 9,714 images across Apple Scab, Black Rot, Cedar Apple Rust, and Healthy classes, achieving 97.8% accuracy, exceeding ResNet-50 ($\sim$92.3%), EfficientNet-B3 ($\sim$94.1%), and MobileNetV3 ($\sim$90.7%) on the same test set while using roughly 4–18$\times$ fewer parameters. Distinctively, the authors integrated Layer-wise Relevance Propagation (LRP) and attention-map visualization to show that the model attends to biologically meaningful regions (scab lesions, blackened margins for black rot, orange rust pustules) rather than background artifacts—directly addressing the "black-box" trust barrier that often limits farmer adoption of deep learning tools. When fine-tuned and externally validated on the six-class Plant Pathology 2021 dataset (including two disease categories never seen during original training), AppleViT retained 95.4% accuracy, evidencing strong transfer capability. The authors additionally benchmarked practical deployment characteristics: TensorFlow Lite conversion succeeded with a $\sim$5 MB quantized model footprint and sub-50 ms inference on a mid-range smartphone CPU, and they explicitly discuss quantization, pruning, and federated learning as next steps toward field-scale deployment.

Together, AppViT and AppleViT indicate that hybrid CNN–ViT and pure lightweight ViT designs can match or exceed the accuracy of substantially larger CNNs at a fraction of the parameter and compute budget—arguably the most consequential trend among the reviewed literature for real-world agricultural deployment.

IX. Contextualizing the Field: Insights from a 45-Study Systematic Review

Doutoum and Tugrul [10] provide a PRISMA-guided systematic review of 45 apple-leaf-disease deep learning studies published between 2016 and 2024, offering useful macro-level context for the case studies above. Key observations include:

- **Publication growth:** annual output rose from a single article in 2016 to 46 in 2023, reflecting rapidly increasing research interest.
- **Dataset composition:** self-constructed datasets were most common ($\sim$36%), followed by PlantVillage, with Plant Pathology and institution-specific datasets used less frequently—reinforcing the fragmentation noted in Section III.
- **Framework usage:** PyTorch was the most frequently used deep learning framework across the surveyed studies, followed by Caffe, MATLAB, and TensorFlow/Keras.
- **Accuracy trend:** a regression across publication year showed a general upward trend in reported accuracy over time, consistent with progressively more sophisticated architectures (attention mechanisms, CBAM modules, hybrid CNN-Transformer designs) and larger training sets.
- **Reported accuracy range:** across the 45 studies, best accuracies ranged from about 63% (Faster R-CNN based detection under complex backgrounds) to 99.8% (capsule/EfficientNet ensembles).

The review's authors emphasize that datasets acquired under single-leaf or laboratory-controlled conditions—the majority of current work—do not reliably predict field performance, and they call for more real-condition, multi-orchard, multi-season datasets and closer researcher–farmer collaboration to ensure practical uptake.

X. Comparative Summary

XI. Common Challenges Identified Across Studies

1) **Dataset fragmentation and realism gap.** Most high-accuracy results (>97%) come from images with plain or lab-controlled backgrounds; performance typically drops when models are exposed to natural orchard scenes with occlusion, variable lighting, and overlapping leaves (as seen in the accuracy gap between Li et al.'s raw vs. lesion-cropped images, and in Liu & Li's more modest 90.7–94.8% detection metrics under complex backgrounds).
2) **Visual similarity between diseases.** Several studies (Li et al.; Al-Wesabi et al.; Gupta & Jadon) report residual confusion between Black Rot and Apple Scab, or between diseased and healthy leaves with subtle early-stage symptoms—a challenge shared across crop species per Li et al.'s cross-species discussion.
3) **The accuracy–efficiency trade-off.** Larger CNNs (VGGNet, ResNet-152) tend to achieve strong accuracy but at prohibitive parameter counts and FLOPs for mobile/edge deployment; the AppViT and AppleViT results suggest hybrid CNN–ViT designs can break this trade-off, but such architectures remain less studied than conventional CNNs.
4) **Limited interpretability.** Most CNN and transfer-learning studies report only aggregate metrics without

TABLE I
COMPARATIVE SUMMARY OF THE REVIEWED STUDIES

Study	Method family	Dataset (approx. size / classes)	Best accuracy	Notable efficiency metric
Li et al. [7]	ACO feature selection + linear SVM	PlantVillage subset, 4 classes	92.5%	Fully interpretable features; no GPU required
Vishnoi et al. [5]	Custom lightweight CNN (3 conv layers)	PlantVillage, 3,171 images, 4 classes	98.0%	Lower storage/time than 9 pretrained CNNs
Ayaz et al. [6]	DCGAN + DCNN	319 real $\rightarrow$ 4,000 synthetic images, 4 classes	99.99%	$\sim$380K–807K parameters
Al-Wesabi et al. [2]	CapsNet + WWO + BiLSTM	$\sim$2,000/class, 4 classes	99.2% (avg. of 5 runs)	2.56 GB memory, 0.518 h training
Liu & Li [4]	Improved YOLOv5 (A-Net: Wise-IoU + RepVGG)	Custom, 12,000 images, 5 diseases + healthy	94.8% mAP@0.5	Object detection (localization), not just classification
Abbaspour-Gilandeh et al. [8]	Sparse coding + shallow ANN	819 fruit images, 18 fine-grained classes	$\sim$90–91% (avg.)	0.175 s inference after dictionary build
Fraiwan et al. [9]	12 pretrained CNNs (transfer learning)	PlantVillage, 3,171 images, 4 classes	$>$99% (all metrics)	Training time varies 119 s–4,240 s across models
Ullah et al. (AppViT) [1]	Hybrid Conv + ViT	Plant Pathology 2021, $\sim$44,500 augmented, 5 classes + healthy	96.4% (precision 0.967)	$\sim$1.3M params, 0.644 GFLOPs, 4.99 MB
Gupta & Jadon (AppleViT) [3]	Lightweight ViT + interpretability (LRP)	9,714 images, 4 classes	97.8%	$\sim$1.3M params, $<$50 ms mobile inference
Doutoum & Tugrul [10]	Systematic review, 45 studies (2016–2024)	Multiple (PlantVillage, Plant Pathology, self-collected)	Range 63–99.8%	Synthesizes frameworks, datasets, trends

explaining *why* a model reached a decision; only AppleViT among the reviewed works incorporates explicit attention/LRP visualization.

5) **Inconsistent evaluation protocols.** Differing train/test splits (50/50 vs. 70/30 vs. 80/20), differing disease taxonomies (3–6 classes), and differing dataset sources make direct cross-study accuracy comparisons only approximately valid.
6) **Small original datasets.** Several works (Ayaz et al.; Abbaspour-Gilandeh et al.) start from only a few hundred original images, relying on augmentation or GAN synthesis to reach viable training-set sizes, raising open questions about generalization beyond the narrow distribution of augmented data.

XII. Future Research Directions

- **Field-validated, multi-season, multi-region datasets** collected under real orchard conditions, ideally shared as open benchmarks to enable fair cross-study comparison.
- **Efficient hybrid CNN–ViT architectures** further optimized via quantization and pruning to push sub-2M-parameter models toward even lower latency on embedded/IoT hardware.
- **Explainable AI as a standard component**—attention visualization, LRP, or Grad-CAM integrated by default into new architectures to support farmer decision-making and regulatory trust.
- **IoT and sensor integration**, combining vision models with environmental sensors (humidity, temperature, leaf wetness) to enable predictive, not just reactive, disease alerts.
- **Detection-and-localization pipelines** that can handle multiple simultaneous lesions per leaf and scale to whole-canopy or drone-based imagery, rather than single cropped-leaf classification.
- **Cross-crop generalization**, extending apple-specific architectures toward unified frameworks applicable to multiple economically important crops.
- **Federated and privacy-preserving learning** to allow multiple orchards/farms to contribute to shared model improvement without centralizing sensitive operational data.

XIII. Conclusion

Across the ten studies reviewed, a clear methodological progression emerges: from handcrafted color/texture/shape features paired with SVMs, through increasingly sophisticated CNN designs (lightweight custom networks, GAN-augmented training, metaheuristic-optimized capsule networks), to modern object-detection frameworks and, most recently, lightweight vision transformers and CNN–ViT hybrids that achieve state-of-the-art accuracy at a fraction of the computational cost of earlier deep CNNs. Reported accuracies frequently exceed 95–99%, but these figures must be interpreted cautiously given the prevalence of curated, background-controlled datasets; performance under natural field conditions, while improving, remains comparatively harder-won. The field's next phase is likely to be defined less by incremental accuracy gains on existing benchmarks and more by validated field deployment, interpretability, and integration into broader precision-agriculture systems.

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